

DEFINITION AND STATISTICAL DISTRIBUTIONS OF A TOPOLOGICAL NUMBER IN THE LATTICE $O(3)$ σ -MODEL*

B. BERG

CERN, Geneva, Switzerland

M. LÜSCHER

Institut für theoretische Physik, Universität Bern, Sidlerstrasse 5, CH-3012 Bern, Switzerland

Received 9 February 1981

We define a topological number Q for spin fields on a two-dimensional lattice. Q assumes integer values only and reduces to the well-known winding number in the classical continuum limit. A Monte Carlo measurement of the topological susceptibility $\chi_t = \langle Q^2 \rangle / \text{volume}$ on a 100×100 lattice reveals that it decreases exponentially with increasing β (= inverse bare coupling constant). The corresponding prediction of the perturbative renormalization group is not matched, however.

1. Introduction

The $O(3)$ σ -model in two space-time dimensions is the simplest asymptotically free [1, 2] field theory. Yet, little has been understood about its dynamics, in particular, the impact of the non-trivial topology of the classical field space remains unclear. In order to make this problem accessible to powerful analytical methods such as the strong coupling expansion, Monte Carlo simulations and the block spin renormalization group, one needs first of all a good definition of the topological charge Q of a lattice spin field.

In the continuum case, the topological charge Q of a three-component spin field $s(x)$, $s(x)^2 = 1$, is defined by

$$Q = \frac{1}{4\pi} \int d^2x s \cdot (\partial_1 s \times \partial_2 s), \quad (1)$$

which is the number of times $s(x)$ winds around the sphere S^2 . A lattice spin field $s(x)$ on the other hand, can be thought of as a net on S^2 with vertices $s(x)$ and rubber bands connecting $s(x)$ with $s(y)$, whenever x and y are nearest neighbors.

* Work supported in part by Schweizerischer Nationalfonds.

This net also winds around S^2 a certain number of times, thus providing a natural definition of the topological charge Q of the field $s(x)$.

In this article we work out an analytic expression for the lattice Q (sect. 2). For completeness, we also include the corresponding formulae for lattice CP^{n-1} fields (sect. 3). We then study the charge distribution in the canonical ensemble defined by the action

$$S = \frac{1}{2}\beta \sum_x \sum_{\mu=1,2} (\partial_\mu s(x))^2. \quad (2)$$

Here, x runs over the vertices of a simple square lattice and ∂_μ denotes the lattice derivative

$$\partial_\mu s(x) = s(x + \hat{\mu}) - s(x), \quad (3)$$

$\hat{\mu}$: unit vector in the positive μ -direction*.

In particular, in sect. 4 we show that the topological susceptibility,

$$\chi_t = \langle Q^2 \rangle / \text{volume}, \quad (4)$$

vanishes to all orders in weak coupling ($\beta \rightarrow \infty$) perturbation theory. From the perturbative renormalization group, the asymptotic behaviour of χ_t is in fact expected to be

$$\chi_t \propto \beta^2 \exp(-4\pi\beta), \quad (\beta \rightarrow \infty). \quad (5)$$

To verify this and the corresponding formula for the magnetic susceptibility [2],

$$\chi_m \propto \beta^{-4} \exp(4\pi\beta), \quad (\beta \rightarrow \infty), \quad (6)$$

we evaluated χ_t and χ_m numerically on a 100×100 lattice using the Monte Carlo method (sect. 5). Whenever they can be compared, our data for the magnetic susceptibility agree with those obtained by other authors [3–5] reporting Monte Carlo simulations of the $O(3)$ σ -model. Although the relatively large lattice allows one to significantly deal with correlation lengths of up to about 30, the scaling laws (5) and (6) are not matched by the Monte Carlo data. Besides summarizing our results, we comment on this discrepancy in sect. 6.

* For convenience, all distances are measured in units of the lattice spacing, in particular, $x = (x_1, x_2) \in \mathbb{Z}^2$.

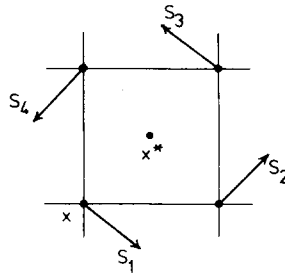


Fig. 1. Drawing of a plaquette with a vertex x^* of the dual lattice at its center. s_1, s_2, s_3 and s_4 are the field vectors attached to the corners of the plaquette.

2. Explicit expression for the topological charge of a lattice spin field

As in the continuum, the charge Q of a lattice spin field $s(x)$ has the local form

$$Q = \sum_{x^*} q(x^*), \quad (7)$$

where the topological density $q(x^*)$ is attached to the vertices x^* of the dual lattice. More precisely, $q(x^*)$ is a function of the four spins s_1, s_2, s_3, s_4 sitting next to x^* (cf. fig. 1):

$$q(x^*) = \frac{1}{4\pi} \{ (\sigma A)(s_1, s_2, s_3) + (\sigma A)(s_1, s_3, s_4) \}. \quad (8)$$

Here, $(\sigma A)(s_1, s_2, s_3)$ denotes the signed area of the spherical triangle with corners s_1, s_2, s_3 . $4\pi Q$ is thus the total signed area of a surface, which is obtained by glueing together all elementary spherical triangles with corners $s_1(x^*), s_2(x^*), s_3(x^*)$ respectively $s_1(x^*), s_3(x^*), s_4(x^*)$ *.

The topological charge Q is not defined for a set of exceptional spin configurations. Namely, when one of the elementary spherical triangles degenerates to a great circle, its area is equal to 2π , but the orientation is ambiguous so that σA is not defined. The set of exceptional spin fields has measure zero in the canonical ensemble. Although not defined for exceptional fields, Q is therefore still measurable and a well-defined observable.

An explicit expression for $(\sigma A)(s_1, s_2, s_3)$ is not difficult to obtain. The area of a spherical triangle with angles $\alpha_1, \alpha_2, \alpha_3$ is

$$A = \alpha_1 + \alpha_2 + \alpha_3 - \pi. \quad (9)$$

* An equally good definition of $q(x^*)$ would be

$$\tilde{q}(x^*) = \frac{1}{4\pi} \{ (\sigma A)(s_1, s_2, s_4) + (\sigma A)(s_2, s_3, s_4) \}.$$

For fields with small action density, $\tilde{q}(x^*) = q(x^*)$, but otherwise $\tilde{q}(x^*) - q(x^*) = 0, \pm 1$.

For $A \neq 0, 2\pi$, its orientation σ is defined by

$$\sigma = \text{sign}[s_1 \cdot (s_2 \times s_3)] = \pm 1. \quad (10)$$

Away from the exceptional configurations

$$s_1 \cdot (s_2 \times s_3) = 0, \quad 1 + s_1 \cdot s_2 + s_2 \cdot s_3 + s_3 \cdot s_1 \leq 0, \quad (11)$$

$(\sigma \cdot A)(s_1, s_2, s_3)$ then extends to a real analytic function taking values

$$-2\pi < A < 2\pi. \quad (12)$$

Explicitly, combining (9) and (10), one finds (after some algebra)

$$\begin{aligned} \exp(\tfrac{1}{2}i\sigma A) &= \rho^{-1} \{1 + s_1 \cdot s_2 + s_2 \cdot s_3 + s_3 \cdot s_1 + is_1 \cdot (s_2 \times s_3)\}, \\ \rho &\stackrel{\text{def}}{=} \{2(1 + s_1 \cdot s_2)(1 + s_2 \cdot s_3)(1 + s_3 \cdot s_1)\}^{1/2} > 0. \end{aligned} \quad (13)$$

Eqs. (7), (8), (12) and (13) thus provide a completely explicit formula for the topological charge Q .

From the geometrical meaning of Q it follows that Q is an integer, provided suitable boundary conditions (e.g. periodic b.c. on an $L \times L$ lattice) are imposed. More important is the topological nature of Q : if the lattice field $s(x)$ is locally deformed in a continuous manner without passing through an exceptional configuration, Q will not change. Indeed, for an infinitesimal deformation $\delta s(x)$, we have

$$\begin{aligned} \delta q(x^*) &= \frac{1}{4\pi} \epsilon_{\mu\nu} \partial_\mu \{ (1 + s(x) \cdot s(x + \hat{\nu}))^{-1} \\ &\quad \times [s(x + \hat{\nu}) \cdot (s(x) \times \delta s(x)) - s(x) \cdot (s(x + \hat{\nu}) \times \delta s(x + \hat{\nu}))] \} \end{aligned} \quad (14)$$

(cf. fig. 1 and sect. 1 for notation) and therefore $\delta Q = 0$, if $\delta s(x) = 0$ outside a bounded region. On the other hand it is quite possible that Q jumps by an arbitrary integer, if the spin field is continuously deformed through an exceptional configuration (such transitions, however, require large angles between some neighboring spins at intermediary stages).

3. Generalization to the lattice CP^{n-1} model*

In a version of the CP^{n-1} model [6], the field variables considered are one-dimensional projection matrices $P(x)$ in an n -dimensional complex space**:

$$P^2 = P, \quad P^+ = P, \quad \text{Tr} P = 1. \quad (15)$$

* This section is not referred to later and may be skipped for a first reading.

** In the notation of ref. [7]: $P_{\alpha\beta} = z_\alpha \bar{z}_\beta$, $|z|^2 = 1$.

Thus, a lattice CP^{n-1} field is an assignment of such a projection matrix to all lattice vertices x . The action is

$$S = \beta \sum_x \sum_{\mu=1,2} \text{Tr} \{ \partial_\mu P(x) \cdot \partial_\mu P(x) \}. \quad (16)$$

For $n = 2$, the CP^{n-1} model is equivalent to the $\text{O}(3)$ model via

$$P(x) = \frac{1}{2}(1 - s(x) \cdot \sigma), \quad \sigma^a: \text{Pauli matrices.} \quad (17)$$

To define the topological charge of a lattice field $P(x)$, we proceed exactly as in sect. 2. The spherical triangles are here replaced by geodesic triangles and the signed area σA by the minimal topological charge $Q(\Delta)$ "enclosed". More precisely, $Q(\Delta)$ is defined as follows. Suppose P_1 , P_2 and P_3 are the corners of a geodesic triangle Δ . Let $\tilde{P}(x)$, $x \in D$, be any differentiable mapping from some region D into CP^{n-1} , such that the boundary ∂D is mapped onto Δ (i.e. $\tilde{P}(x)$ represents a two-dimensional surface in CP^{n-1} with boundary Δ). Then $Q(\Delta) \in (-\frac{1}{2}, \frac{1}{2}]$ is given by

$$Q(\Delta) = \frac{i}{2\pi} \int_D d^2x \epsilon_{\mu\nu} \text{Tr} \{ \tilde{P} \partial_\mu \tilde{P} \partial_\nu \tilde{P} \} (\text{mod } \mathbb{Z}). \quad (18)$$

Note that the integral is the topological charge carried by $\tilde{P}$. Two different choices $\tilde{P}$ and $\tilde{P}'$ carry the same charge (modulo integers) so that $Q(\Delta)$ is well-defined. Explicitly, we have

$$\exp 2\pi i Q(\Delta) = \text{Tr} \{ P_3 P_2 P_1 \} \{ \text{Tr} P_1 P_2 \text{Tr} P_2 P_3 \text{Tr} P_3 P_1 \}^{-1/2}. \quad (19)$$

Thus, to obtain the topological charge Q of a lattice CP^{n-1} field, one merely has to replace σA in eq. (8) by $4\pi Q(\Delta)$ as given above [eq. (19)].

4. Vanishing of the topological susceptibility in weak coupling perturbation theory

The topological susceptibility χ_t is defined by

$$\chi_t = \sum_{x^*} \langle q(x^*) q(y^*) \rangle, \quad (20)$$

where $\langle \dots \rangle$ denotes the expectation value

$$\langle \dots \rangle = \frac{1}{Z} \int \prod_x d^3s(x) \delta(s(x)^2 - 1) \dots \exp - S, \quad (21)$$

and the action S is given by eq. (2). In a finite volume with periodic boundary conditions, eq. (20) is equivalent to eq. (4). In weak coupling perturbation theory one

only integrates over fields $s(x)$, which are *infinitesimally* close to a constant field. In a finite volume, $Q = 0$ in a whole neighborhood of the constant fields and therefore, by eq. (4), $\chi_t = 0$ to all orders in β^{-1} .

To show that the same conclusion holds in the infinite volume case, too, requires a little detour. First of all note that to any finite order in β^{-1} , the complicated expression [eqs. (8), (13)] for $q(x^*)$ can actually be replaced by an $O(3)$ invariant polynomial of s_1, s_2, s_3 and s_4 , because

$$s_i \cdot s_j = 1 + O(\beta^{-1}), \quad \sigma A = O(\beta^{-1}). \quad (22)$$

For example,

$$q(x^*) = \frac{1}{8\pi} \{ [s \cdot (\partial_1^* s \times \partial_2 s)](x + \hat{1}) + [s \cdot (\partial_1 s \times \partial_2^* s)](x + \hat{2}) \} + O(\beta^{-2}), \quad (23)$$

where ∂_μ^* is the adjoint of ∂_μ :

$$\partial_\mu^* s(x) = s(x) - s(x - \hat{\mu}). \quad (24)$$

By Elitzur's theorem [8] (which was proved recently by David [9]), it follows that the two-point function $\langle q(x^*)q(y^*) \rangle$ is infrared finite to all orders in β^{-1} . Furthermore, at any finite order in β^{-1}

$$\langle q(x^*)q(y^*) \rangle \sim |x - y|^{-4}, \quad (|x - y| \rightarrow \infty), \quad (25)$$

modulo logarithmic corrections, because $q(x^*)$ has dimension two. The perturbative expansion of χ_t is therefore well-defined and, order by order,

$$\chi_t = \lim_{V \rightarrow \infty} \chi_t^V, \quad (26)$$

where χ_t^V denotes the topological susceptibility in a finite volume V . Hence $\chi_t = 0$ to all orders of β^{-1} .

The vanishing of χ_t in perturbation theory suggests that χ_t decreases exponentially for large β . In fact, one expects that in the continuum limit

$$\chi_t \sim (\text{cutoff})^{-2} \times m^2, \quad (27)$$

where m is a renormalization group invariant mass. At least this is what one finds in the large- n limit of the CP^{n-1} model [7]. Accordingly, χ_t should scale as in eq. (5) for large β [2] (and infinite volume, of course).

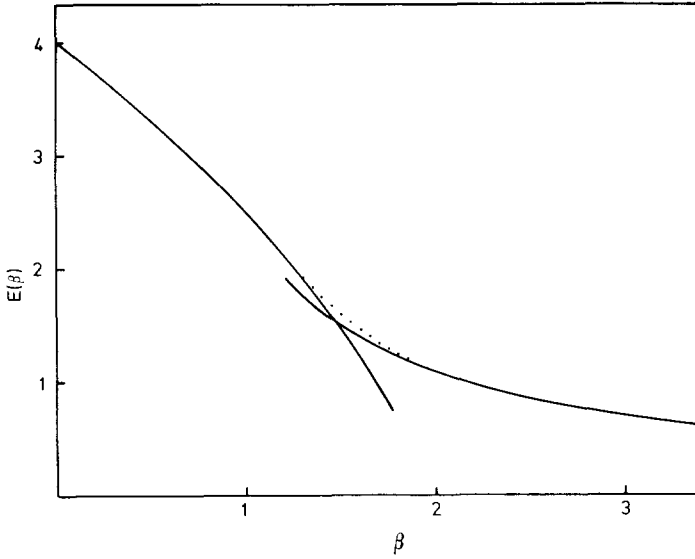


Fig. 2. Plot of the energy E [eq. (29)] of the $O(3)$ σ -model. Dots denote Monte Carlo points and the curves represent the strong respectively weak coupling series (30) and (31).

5. Monte Carlo simulation of the $O(3)$ σ -model

In this section, we report the results of a numerical evaluation of χ_t and the magnetic susceptibility

$$\chi_m \stackrel{\text{def}}{=} \sum_x \langle s(x) \cdot s(y) \rangle \quad (28)$$

on a 100×100 lattice with periodic boundary conditions. We used the heat bath version of the Monte Carlo method [10], which is described in detail for lattice gauge theories in ref. [11]. To check our programs, we have first calculated the internal energy

$$E \stackrel{\text{def}}{=} \sum_{\mu=1,2} \langle \partial_\mu s(y) \cdot \partial_\mu s(y) \rangle. \quad (29)$$

This is shown in fig. 2 together with the strong and weak coupling expansions

$$E = 4 - 4y - 8y^3 - \frac{48}{5}y^5, \quad y = \frac{\text{ch}\beta}{\text{sh}\beta} - \frac{1}{\beta} = \frac{1}{3}\beta + \dots, \quad (30)$$

$$E = \frac{2}{\beta} + \frac{1}{4\beta^2} + 0.156 \frac{1}{\beta^3}. \quad (31)$$

TABLE 1
 Monte Carlo results for the energy (29), the magnetic susceptibility (28)
 and the topological susceptibility (20) in the $O(3)$ σ -model on
 a 100×100 lattice

β	E	χ_m	$10^4 \cdot \chi_t = \langle Q^2 \rangle$
1.1	2.295	14.4 ± 0.4	125 ± 5
1.2	2.109	23.5 ± 0.8	98 ± 4
1.3	1.924	39 ± 3	64.7 ± 1.8
1.4	1.751	82 ± 8	41.3 ± 1.3
1.5	1.592	191 ± 21	23.7 ± 0.9
1.6	1.457	370 ± 48	12.8 ± 0.7
1.7	1.342	871 ± 61	5.7 ± 0.3
1.8	1.247	1634 ± 82	2.7 ± 0.2
1.9	1.167	2285 ± 58	1.05 ± 0.06
2.0	1.098	2720 ± 85	0.52 ± 0.03

Only in the interval $1.3 < \beta < 1.9$ are the data visibly off the analytic curves. We have not seen any sign of a hysteresis in this “crossover region”.

More interesting are the results for χ_m and χ_t , which are summarized in table 1. For each value of β we made 8100 sweeps through the lattice plus a number of sweeps (thousands at times) to establish equilibrium. E and χ_m were measured for all sweeps, but since the evaluation of Q for a given spin configuration is time consuming, only the last 2100 sweeps were used to compute $\langle Q^2 \rangle$. The statistical errors were estimated by dividing the sweeps into bunches of 300, measuring the standard deviation of the bunch averages from the total average and dividing the square root of the number of bunches.

From the table we see that χ_m increases exponentially in the region $1.3 \lesssim \beta \lesssim 1.7$. The correlation length varies here from about 4 to 30 [3], i.e. in this range neither finite cutoff nor finite volume effects dominate. It is not unreasonable therefore, to expect χ_m to scale in this interval as predicted by the perturbative renormalization group [eq. (6)]. As shown in fig. 3, however, this is not the case. Rather than approaching a constant, the quantity

$$\delta_m = 2 \cdot 10^5 \beta^4 e^{-4\pi\beta} \chi_m \quad (32)$$

decays exponentially, corresponding to

$$\chi_m \sim \beta^{-4} \exp 3.3\pi\beta \quad (33)$$

This amounts to a change of δ_m by about a factor $\frac{1}{2.5}$ when going from $\beta = 1.3$ to 1.7. The solid lines in fig. 3 are the strong and weak coupling curves obtained from the tenth-order series produced by Camp and Van Dyke [12] from

$$\chi_m = 10^4 \{ 1 - 1.5632\beta^{-1} + 0.23176\beta^{-2} \}. \quad (34)$$

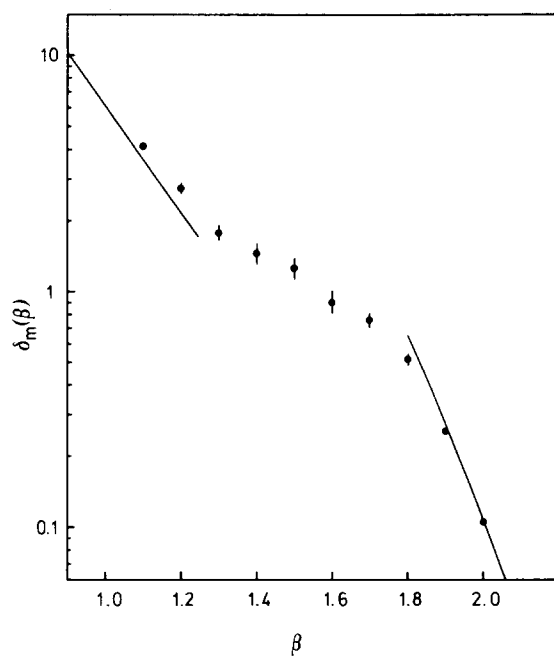


Fig. 3. Plot of the Monte Carlo data for the magnetic defect (32) in the $O(3)$ σ -model on a 100×100 lattice.

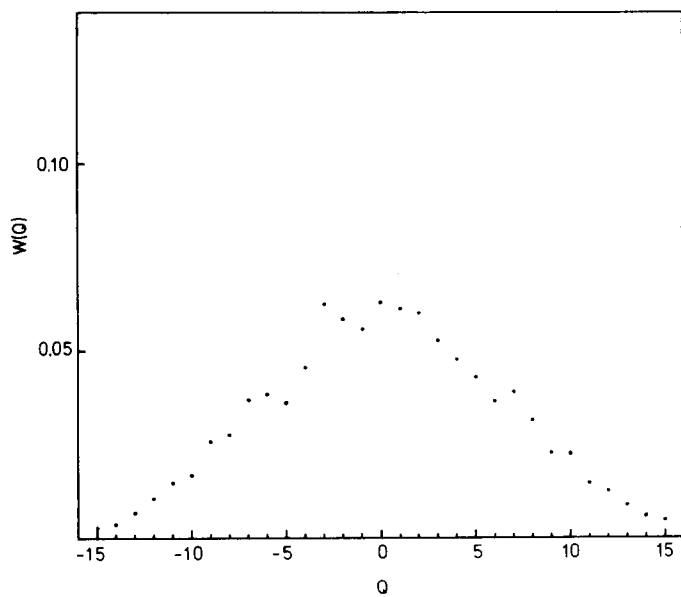


Fig. 4. Probability $W(Q)$ to find a topological charge Q at $\beta = 1.4$.

Eq. (34) is the result of a finite volume spin wave calculation and involves the numerical evaluation of some one-loop Feynman diagrams.

Even more dramatic is the deviation from the scaling law (5) for the topological susceptibility. The charge distributions at $\beta = 1.4$ and 1.6 are shown in figs. 4 and 5, respectively, and the topological defect

$$\delta_t = 10^{-5} \beta^{-2} e^{4\pi\beta} \chi_t \quad (35)$$

is displayed in fig. 6. δ_t apparently increases exponentially in the scaling region by about a factor 8 from $\beta = 1.3$ to 1.7. A simple fit of the data for $1.4 \leq \beta \leq 1.7$ gives

$$\chi_t \sim \beta^{-2} \exp(-2.5\pi\beta). \quad (36)$$

The solid line also shown in fig. 6 is obtained from the strong coupling limit

$$\chi_t = \frac{1}{16}, \quad (\beta = 0). \quad (37)$$

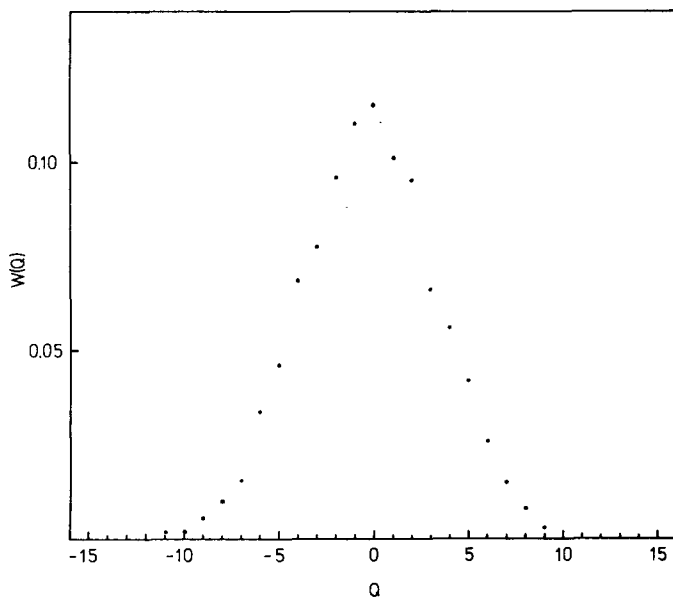


Fig. 5. Same as fig. 4, but at $\beta = 1.6$.

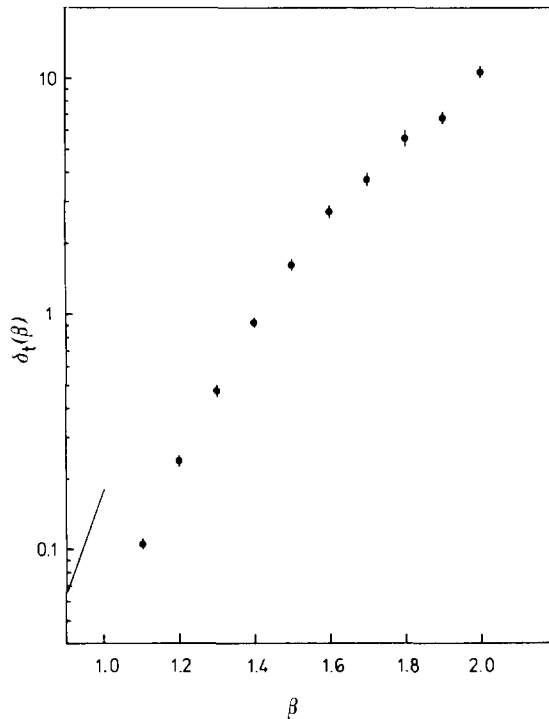


Fig. 6. Plot of the Monte Carlo data for the topological defect (35) in the $O(3)$ σ -model on a 100×100 lattice.

6. Conclusions

From a purely geometrical point of view, our definition of the topological charge Q seems most natural. It also has the characteristic property that the topological susceptibility χ_t vanishes to all orders in perturbation theory. Furthermore, the analytic form of Q is sufficiently simple to allow strong coupling, numerical and other methods to be applied to determine e.g. statistical distributions of Q .

The results of the Monte Carlo simulation of the $O(3)$ σ -model on a 100×100 lattice show a region $1.3 \lesssim \beta \lesssim 1.7$ where χ_m and χ_t scale exponentially. However, the data are definitely off the curves predicted by the perturbative renormalization group. This is rather surprising in view of the large correlation length ξ varying from about 4 to 30. For comparison we have calculated the defect

$$\delta_m = 32\beta e^{-4\pi\beta} \chi_m \quad (38)$$

in the large- n limit of the $O(n)$ σ -model on a 100×100 lattice (see fig. 7). The scaling region here is about $0.5 \lesssim \beta \lesssim 0.7$, which corresponds to roughly $4 \lesssim \xi \lesssim 20$.

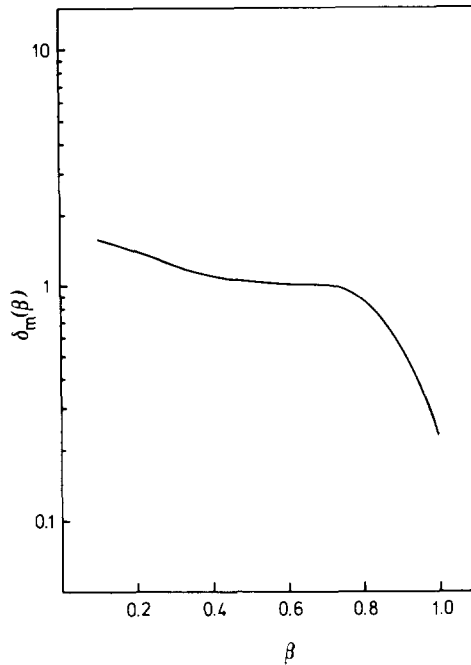


Fig. 7. Plot of the magnetic defect in the $O(n)$ σ -model on a 100×100 lattice in the large- n limit.

From fig. 7 we see that δ_m is close to 1 in the whole scaling region, i.e. the situation looks rather different from what we found in the $O(3)$ case.

Studying the magnetic susceptibility χ_m , Martinelli et al. [5] also observed that the data were not fitted by the two-loop renormalization group formula (6). They suggest that the discrepancy may be due to large perturbative corrections to eq. (6). This is quite possible although the size of the deviation can hardly be explained with the first order correction alone.

In the case of the topological susceptibility on the other hand, the deviation from the renormalization group formula (5) is so large that a huge power of β would be needed to correct for it, i.e. the effect is more likely to persist, when $\beta \rightarrow \infty$. If true, it means that one of our assumptions was not justified. We do not know for sure which one, but there are at least three possibilities.

(a) Our lattice topological charge Q may not approximate the continuum topological charge sufficiently well. One should be able to check this in perturbation theory and in the large- n limit of the lattice CP^{n-1} model as defined in sect. 3.

(b) Provided the continuum limit of the lattice σ -model exists, the anomalous scaling of χ_t means that there is an unexpected ultraviolet divergence, when one tries to define the topological susceptibility

$$\chi_t^c = \int d^2x \langle q(x)q(0) \rangle, \quad q: \text{topological density} \quad (39)$$

in the continuum theory. At first sight, one might think that this divergence is related to the infinite density of the exact instanton gas [13, 14], which results from an ultraviolet divergent instanton scale size integral (see ref. [15] for a summary). However, this divergence is only logarithmic, whereas, by eq. (36), χ_1^c is expected to diverge with a power $p \sim \frac{1}{2}$ of the cutoff.

(c) A more radical possibility is that the lattice σ -model considered here does not have a decent continuum limit for $\beta \rightarrow \infty$ (although the correlation length may very well scale exponentially).

At any rate, it is important to look for a convincing explanation of our Monte Carlo data, for otherwise one could not be sure that the same does not happen in lattice gauge theories, for example.

We are indebted to P. Schwab for writing a program to plot spin fields. We also thank him, J. Gasser and K. Amsler for general help while operating the computer.

References

- [1] A.M. Polyakov, Phys. Lett. B59 (1975) 79
- [2] E. Brézin and B. Zinn-Justin, Phys. Rev. B14 (1976) 3110
- [3] S.H. Shenker and J. Tobochnik, Phys. Rev. B22 (1980) 4462
- [4] C.P. Korthals-Altes, Talk at Symp. on Topical questions in QCD, Copenhagen (June, 1980); G. Immirzi, Harvard preprint HUTMP 80/B102 (November, 1980)
- [5] G. Martinelli, G. Parisi and R. Petronzio, CERN preprint (December, 1980)
- [6] H. Eichenherr, Nucl. Phys. B146 (1978) 215; (E: B155 (1979) 544); V. Golo and A. Perelomov, Phys. Lett. 79B (1978) 112
- [7] A. D'Adda, P. Di Vecchia and M. Lüscher, Nucl. Phys. B146 (1978) 63
- [8] S. Elitzur, Institute for Advanced Study preprint (1979) unpublished
- [9] F. David, Saclay preprint DPh.T. 64/80 (1980)
- [10] N. Metropolis, A.W. Rosenbluth, A.H. Teller and E. Teller, J. Chem. Phys. 21 (1953) 1087
- [11] M. Creutz, Phys. Rev. D21 (1980) 2308
- [12] W.J. Camp and J.P. Van Dyke, J. Phys. C8 (1975) 336
- [13] V.A. Fateev, I.V. Frolov and A.S. Schwarz, Nucl. Phys. B154 (1979) 1
- [14] B. Berg and M. Lüscher, Comm. Math. Phys. 69 (1979) 57
- [15] M. Lüscher: Exact instanton gases, in Recent developments in gauge theories, ed. G. 't Hooft et al. (Cargèse 1979), (Plenum Press, New York, 1980)